Analytical report

Introduction and data

Question 1. Understanding and preparing the data.

The UCI Student Performance dataset contains data on student achievement in secondary education at two Portuguese schools. This dataset originates from a study by Cortez (2008), where the data was collected using school reports and questionnaires. The dataset is available through the UCI Machine Learning Repository. This report will focus on the dataset on student performance in Mathematics course.

Our main research question is which student characteristics are associated with final Mathematics performance, and which factors are linked to the risk of failing. We will be discussing what reasonable conclusions can be derived from the data, whether we should be concerned about uncertainty, and where the limitations of the data and the report are.

After importing the dataset, we can see that there are 33 columns and 395 rows. Each row represents a student, and each column is a variable. The data attributes include student grades, demographic (e.g., age), social and school-related variables (e.g., school absences). For convenience, all variables and their characteristics are organized in Table 1, Appendix A.

The main outcome variable is the final Mathematics grade (see variable “G3”), which is measured on a numerical scale from 0 to 20. As a numerical variable, it can be useful as a numerical outcome in a linear regression analysis to predict the student's final grade. The second outcome variable (see variable “pass”) was not originally part of the dataset and was created for the purposes of the project. The binary variable indicates whether a student passed or failed the final course, with pass defined as having a final grade of 10 or higher. This variable can be useful in a classification task using a logistic regression model to estimate the probability of passing.

The predictor variables include the students' demographic characteristics, family background, school-related factors, study habits, social activities, and health-related information. These variables can be classified into three types:

- Numerical variables (e.g., “age”, “absences”),
- Categorical variables (e.g., “sex”, “school”),
- Ordinal variables (e.g., “studytime”, “failures”)

While exploring the dataset, we checked for any important data-quality or data-use issues. Regarding data quality, there are no missing values and no duplicate rows in the dataset. Although, some unusual observations, particularly the wide range of absences, should be investigated to determine whether they represent a valid observation or a data-entry error. In addition, some variables are categorical even though their values are represented numerically. Understandably, categorical variables will be encoded as factors in R and interpreted appropriately.

In terms of data-use issues, we should consider the generalizability of the findings since the dataset includes information from only two Portuguese schools. The findings may not be applicable to other schools, educational systems, or student populations. It is notable that the distribution across the two schools is uneven, which may also limit the reliability of school comparisons and the generalizability of the findings. Another important consideration is target leakage. In order to avoid target leakage, we should not use the final grade as a predictor when

predicting the second outcome variable, since the second variable derives from the first. At last, the strong correlation between first-period grade (G1), second-period grade (G2), and final grade may lead the model to use previous grades to predict the final grade, which is not helpful in answering our research question. Therefore, we do not use previous grades as predictors in the main regression model.

As a last thought, privacy, potential bias, and the distinction between correlation and causation must be considered. Predictive models should be interpreted cautiously and should not be used as the sole basis for decisions affecting students' educational opportunities. A potential overlap between predictors and multicollinearity may reduce the precision of regression coefficients and make individual associations more difficult to interpret. Since the dataset is observational, the identified associations should not be interpreted as evidence of causal relationships.

Exploratory analysis

Question 2. Exploring the data.

Descriptive statistics were calculated for 395 students from two Portuguese schools, Gabriel Pereira and Mousinho da Silveira. The sample is unevenly distributed across schools, with students from Gabriel Pereira accounting for 88.35% of the total sample. The majority of students, 208 (52.66%), were female. The mean age of the students was 16.69 (*SD* = 1.27, *Median* = 17, IQR = 2), with ages ranging from age 15 to 22. A desire to pursue higher education was expressed by 375 students (94.93%), indicating that the sample was heavily skewed towards students with this aspiration. The average number of absences was 5.71 (*SD* = 8.00), while the median was 4. Absences ranged from 0 to 75, with an IQR of 8. The mean being higher than the median, together with the maximum value of 75, suggests a positively skewed distribution, in which a small number of students may have had substantially more absences than others. The median study-time category was 2, corresponding to 2–5 hours of studying per week. Responses ranged from under 2 hours (study-time category 1) to over 10 hours (study-time category 4).

Regarding the final Mathematics grade, the mean final grade was 10.41 (*SD* = 4.58), while the median was 11. Final grades ranged from 0 to 20, with an IQR of 6. Figure 1 presents the distribution of the final grade of students, which is unimodal and slightly negatively skewed (Appendix B). Out of 395 students, 265 scored 10 or higher and passed (67.09%), while 130 failed (32.91%).

Study time, educational aspirations, school, and sex are potentially relevant variables for examining differences in student performance. The median final grade is higher in study-time category 3, which represents students that spend 5 to 10 hours studying weekly (Figure 2, Appendix B). Categories 3 and 4 have similar median grades. This suggests a possible positive association between weekly study-time and final grade in a Mathematics course, although the distributions overlap. Students with an intention to continue their education in higher educational institutions have a higher median final grade than students who do not (Figure 3, Appendix B). This could imply a positive association between desire for higher education and academic performance. These descriptive comparisons cannot establish that studying more or wanting to pursue higher education increases final Mathematics grade.

Regarding differences by sex, male students had a slightly higher pass percentage than female students (Figure 4, Appendix B). The pass rate was also slightly higher among students at Gabriel Pereira than among those at Mousinho da Silveira (Figure 5, Appendix B). When pass

rates were compared jointly by sex and school (Figure 6, Appendix B), male students at Gabriel Pereira had the highest pass rate, while male students at the Mousinho da Silveira had the lowest pass rate. Female students at the same school had higher pass rate when compared with the female students at the Gabriel Pereira. These differences describe the observed patterns in the sample but do not establish that sex or school accounts for the differences in pass rates.

Overall, the descriptive findings highlight variation in Mathematics performance in relation to study time, educational aspirations, sex, and school. The uneven school representation, the imbalance in educational aspirations, and the unusually high maximum number of absences are important considerations for subsequent analyses. Further statistical testing would be required to determine whether the observed group differences represent statistically significant associations.

Relationships between variables

Question 3. Investigating relationships.

The correlation analysis indicated that the final grades were most strongly associated with the number of past class failures, with a correlation coefficient of $r = -0.36$. This suggests a weak-to-moderate negative linear relationship. Further associations were observed between grades and the education levels of the parents, with correlation coefficients of $r = 0.21$ (Medu – mother's education level) and $r = 0.15$ (Fedu – father's education level). This implies weak positive linear relationships. Weak negative correlations were observed between grades and age ($r = -0.16$), frequency of going out with friends ($r = -0.13$), and travel time from home to school ($r = -0.11$). In contrast, weekly study time, quality of family relationships, number of school absences, free time after school, health status, and alcohol consumption showed very weak linear correlations with final grades. These findings should be interpreted cautiously because some predictors represent ordered categories (e.g., parental education and previous class failures) rather than continuous measurements. Additional statistical tests are needed to determine whether the observed associations are statistically significant.

The contingency table examined the association between the intention to pursue higher education and pass/fail status. Students who intended to pursue higher education had a higher pass rate (68.8%) than those who did not (35%). This suggests an association between the desire to pursue higher education and academic performance. A second contingency table examined the association between extra educational support and pass/fail status. Most students reported that they did not receive extra educational support (N = 344). Among students who received support, 54.9% passed, compared with 68.9% of those who did not receive it. It is important to note that the pass rate was lower among students who received extra educational support. This finding should be interpreted cautiously, since students experiencing academic difficulties may be more likely to receive extra support from their school. Further testing, such as a chi-square test of independence, would be needed to determine whether these associations are statistically significant.

The distribution of final grades across the two schools shows that students from Gabriel Pereira had a slightly higher median final grade, suggesting generally better academic performance (Figure 7, Appendix B). The spread of final grades was also somewhat wider, indicating greater variability in student performance. In contrast, Mousinho da Silveira had a lower median grade and one notably low outlier. Both distributions overlapped, suggesting variation in grades within both schools.

The distribution of final grades by the number of previous class failures shows that the highest median final grade was observed among students with no previous failures (Figure 8, Appendix B). The median decreased as the number of failures increased. Students with one or more previous failures also showed variability in their grades. Overall, the figure suggests a negative association between past class failures and final academic performance.

Based on the exploratory findings, the number of past failures is the most promising predictor of final grades because it showed the strongest observed correlation. The desire to pursue higher education and parental education also appeared relevant. School and sex could be considered additional predictors to examine potential group differences. These variables provide a basis for selecting predictors for the regression models. The exploratory findings alone do not establish statistical significance, causality, or independent predictive value. The final selection of predictors should therefore also consider the regression results, potential overlap between predictors, and the interpretability of the models.

Question 4. Building and interpreting statistical models.

A multiple linear regression model was fitted to examine the association the final grades and student characteristics. The predictors included weekly study time, number of previous failures, number of school absences, sex and intention to pursue higher education. Earlier grades were excluded from this model to focus on student characteristics. The overall model was statistically significant, F (5.389) = 15.2, $p < 0.001$, explaining 16.4% of the variance in final grades ($R^2 = 0.164$, Adjusted $R^2 = 0.153$).

Previous class failures showed the clearest association with final grades ($b = -2.02$, $p < 0.001$). Each additional failure was associated with an estimated 2.02-point decrease in the final grade, when holding other predictors constant. Sex was also significantly associated with academic performance ($b = 1.48$, $p < 0.001$). This means that male students had an estimated final grade 1.48 points higher than female students. Finally, the desire to pursue higher education was positively associated with final grades ($b = 2.05$, $p = 0.04$), which implies that students with an intention to pursue higher education had an estimated 2.05-point advantage over those who did not intend.

Study time ($b = 0.42$, $p = 0.11$) and number of school absences ($b = 0.04$, $p = 0.10$) were not statistically significant. The findings above demonstrate insufficient evidence of independent associations in this model, but they also do not establish that the variables are unrelated to grades. The multiple linear regression model identified previous class failures as an important correlate of final grades, while having modest exploratory power and substantial variation remaining unexplained. Sex and desire to pursue higher education were also significantly associated with academic performance.

A logistic regression model was also fitted to examine the relationship between the same predictors and the binary pass/fail outcome. Previous class failures ($OR = 0.39$, $p < 0.001$) and sex ($OR = 1.65$, $p = 0.04$) were statistically significant in this reported output, whereas weekly study time, number of school absences, and desire to pursue higher education were not statistically significant at the 5% level. When controlling for the other predictors, each additional previous failure was associated with approximately 60.1% lower odds of passing. Male students had estimated odds of passing approximately 65% higher than female students. Overall, the model's residual deviance (449.53) was lower than its null deviance (500.50), suggesting improved fit compared with an intercept-only model.

The multiple linear regression model estimates associations with numerical final grades, while the logistic regression estimates associations with the probability of meeting the pass threshold. Therefore, the linear model provides information on the differences in grade points, and the logistic regression gives a complementary look on whether student characteristics are associated with passing. Both models are equally important in understanding the patterns in student performance.

Model evaluation and uncertainty

Question 5. Evaluating the analysis.

A residuals-versus-fitted plot for the linear regression was inspected to assess whether the linear model showed obvious patterns and comment on any clear warning signs (Figure 9, Appendix B). It showed that the residuals are not randomly distributed around zero and the blue smooth curves illustrate a noticeable pattern. The curve remains relatively stable in the middle, but at lower fitted values it decreases and at the higher fitted values it falls. Thus, this may indicate that the linear regression is not capturing fully the relationship between variables. Finally, some observations in the plot, like the spread of residuals which varies across fitted values and some cases of relatively large residuals, suggest that the assumptions of linearity and constant variance may not be fully satisfied.

Uncertainty in the linear regression estimates was assessed using 95% confidence intervals and p-values. Previous class failures showed a clear negative association with final grades ($b = -2.03$, 95% *CI* [−2.62, −1.43], $p < .001$), while male students had higher estimated grades than female students, holding the other predictors constant ($b = 1.48$, 95% *CI* [0.60, 2.37], $p = .001$). Intention to pursue higher education was also positively associated with grades, although the evidence was weaker ($b = 2.06$, 95% *CI* [0.03, 4.09], $p = .046$). In contrast, the confidence intervals for study time and absences included zero, indicating uncertainty about their independent associations with final grades. Although some coefficients were statistically significant, this does not establish causality or practical importance. The findings should also be considered alongside the model's modest explanatory power and the possibility of overlapping between predictors.

Uncertainty in the logistic regression estimates was assessed using 95% confidence intervals and p-values. Previous class failures were significantly associated with lower odds of passing ($OR = 0.399$, 95% *CI* [0.278, 0.553], $p < .001$), with each additional failure associated with approximately 60.1% lower odds of passing, holding the other predictors constant. Sex was also statistically significant ($OR = 1.65$, 95% *CI* [1.02, 2.71], $p = .043$), although the confidence interval was relatively wide and only narrowly excluded 1. In contrast, study time, absences, and intention to pursue higher education were not statistically significant at the 5% level. In particular, the wide confidence interval for higher-education intention ($OR = 2.18$, 95% *CI* [0.743, 6.62], $p = .157$) indicates substantial uncertainty about the size and direction of its independent association with passing.

The two regression models provide the clearest evidence of an association between previous class failures and academic performance. However, statistical importance does not equate practical importance, and neither model establishes causality. The modest exploratory power of the linear model limits the confidence in the precise size of the effects, alongside with uncertainty in several coefficients, possible overlap between predictors, and the sample's uneven representation of the two schools. The linear model's residuals-versus-fitted plot raised issues about non-linearity, unequal residual variance, unusual observations, and unsatisfied assumptions.

Overall, the results identify useful patterns, but their predictive performance and generalizability should not be overstated.

Causality, limitations, and conclusions

Question 6. Drawing appropriate conclusions.

The data are observational since there was no experimental manipulation (e.g., students being randomly assigned to different levels of study time, previous class failures, or other characteristics). Consequently, the regression models identify associations rather than causal relationships. It is not possible, for instance, to prove that failures themselves cause lower grades, because of the negative association between previous class failures and academic performance. Other factors, such as prior academic difficulties, learning needs, family circumstances, motivation, or access to educational support, may contribute to both previous failures and current performance.

There are some potential data-quality issues and measurement limitations that should be a matter of concern. Most variables were recorded based on reported information. Self-reported measures may be affected by recall errors or differences in how students interpret the questions. In addition, variables included in the models may not capture all relevant influences on academic performance, such as teaching quality, socioeconomic circumstances, or differences in school resources. The models therefore may be affected by omitted-variable bias. The coding of variables and the possibility of unusual observations should be considered when interpreting the estimates.

The imbalance in the sampling, with students from Gabriel Pereira representing 88.35% of the sample, may lead to findings that reflect mostly the characteristics of the more heavily represented school and not adequately represent students from Mousinho da Silveira. Additionally, results from two schools in one country cannot automatically be generalized to other schools, countries, or educational systems, because curricula, assessment practices, student populations, and access to educational resources may differ.

Across both models, the clearest and most consistent association was observed between academic performance and past class failures. Each additional previous failure was associated with an estimated decrease of approximately 2.03 points in final grade and approximately 60.1% lower odds of passing, holding the other included predictors constant. Sex was also statistically significant in both models, with male students having higher estimated grades and odds of passing than female students in this sample, after adjustment for the other predictors. Intention to pursue higher education was positively associated with final grades in the linear model, but its association with passing was not statistically significant in the logistic model. Study time and absences were not statistically significant predictors in either model at the 5% level.

The analysis suggests that previous class failures are an important indicator of subsequent academic performance within this sample. The models, although, explain only a modest proportion of the variation in the final grades, and several estimates remain uncertain. The results do not establish causality, and the conclusion should therefore be treated as evidence of associations in this dataset rather than universal explanations of student performance.

Reference List

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Appendix A

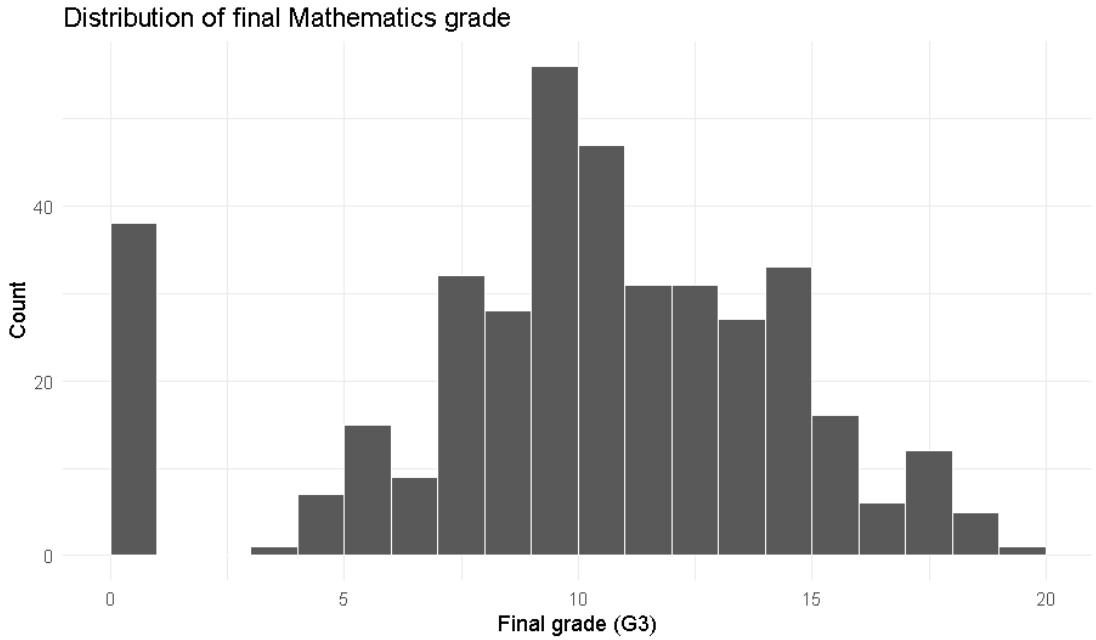
Table 1.

No.	Variable	Description	Type	Values
1	school	Student's school	Binary	GP (Gabriel Pereira), MS (Mousinho da Silveira)
2	sex	Student's sex	Binary	F (female), M (male)
3	age	Student's age	Numeric	15–22 years
4	address	Type of home address	Binary	U (urban), R (rural)
5	famsize	Family size	Binary	LE3 (3 or fewer), GT3 (more than 3)
6	Pstatus	Parents' cohabitation status	Binary	T (living together), A (living apart)
7	Medu	Mother's education level	Ordinal	0 (none), 1 (primary education), 2 (5th–9th grade), 3 (secondary education), 4 (higher education)
8	Fedu	Father's education level	Ordinal	0 (none), 1 (primary education), 2 (5th–9th grade), 3 (secondary education), 4 (higher education)
9	Mjob	Mother's occupation	Nominal	teacher, health, services, at_home, other
10	Fjob	Father's occupation	Nominal	teacher, health, services, at_home, other
11	reason	Reason for choosing the school	Nominal	home, reputation, course, other
12	guardian	Student's guardian	Nominal	mother, father, other
13	traveltime	Travel time from home to school	Ordinal	1 (under 15 min), 2 (15–30 min), 3 (30 min–1 hour), 4 (over 1 hour)

14	studytime	Weekly study time	Ordinal	1 (under 2 hours), 2 (2–5 hours), 3 (5–10 hours), 4 (over 10 hours)
15	failures	Number of past class failures	Numeric	1–3 for the corresponding number of failures; 4 otherwise
16	schoolsup	Extra educational support	Binary	yes, no
17	famsup	Family educational support	Binary	yes, no
18	paid	Extra paid classes in the course subject	Binary	yes, no
19	activities	Participation in extracurricular activities	Binary:	yes, no
20	nursery	Attended nursery school	Binary	yes, no
21	higher	Desire to pursue higher education	Binary	yes, no
22	internet	Internet access at home	Binary	yes, no
23	romantic	Currently in a romantic relationship	Binary	yes, no
24	famrel	Quality of family relationships	Ordinal	1 (very bad)–5 (excellent)
25	freetime	Free time after school	Ordinal	1 (very low)–5 (very high)
26	goout	Frequency of going out with friends	Ordinal	1 (very low)–5 (very high)
27	Dalc	Workday alcohol consumption	Ordinal	1 (very low)–5 (very high)
28	Walc	Weekend alcohol consumption	Ordinal	1 (very low)–5 (very high)
29	health	Current health status	Ordinal	1 (very bad)–5 (very good)
30	absences	Number of school absences	Numeric	0–93
31	G1	First-period grade	Numeric	0–20
32	G2	Second-period grade	Numeric	0–20
33	G3	Final grade (output target)	Numeric	0–20
34	pass	Outcome of final grade	Binary	Pass, Fail

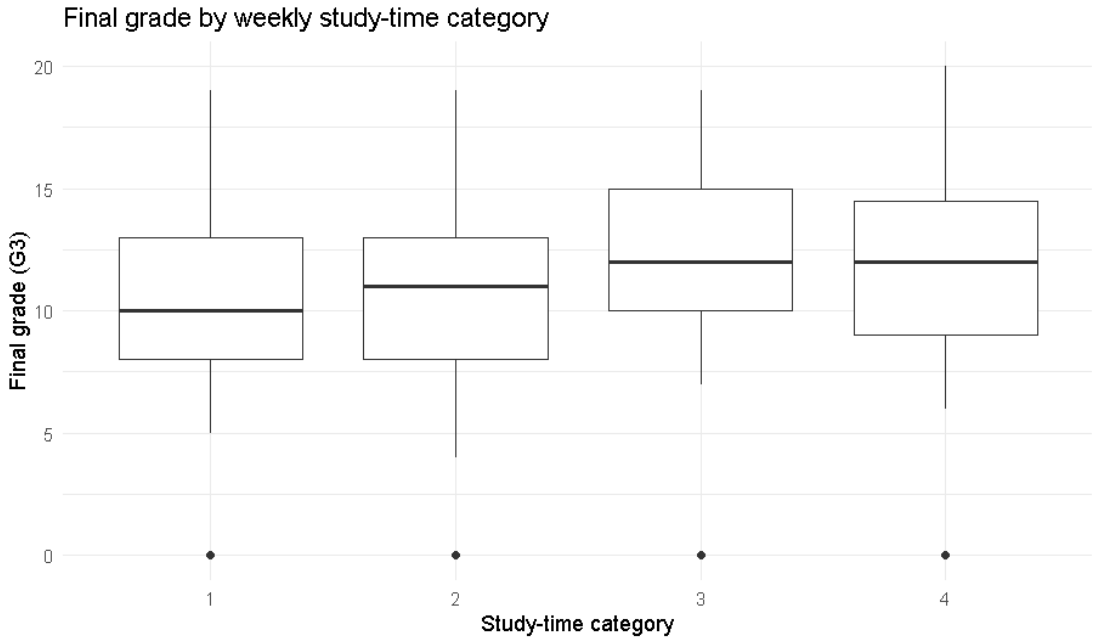
Appendix B

Figure 1.



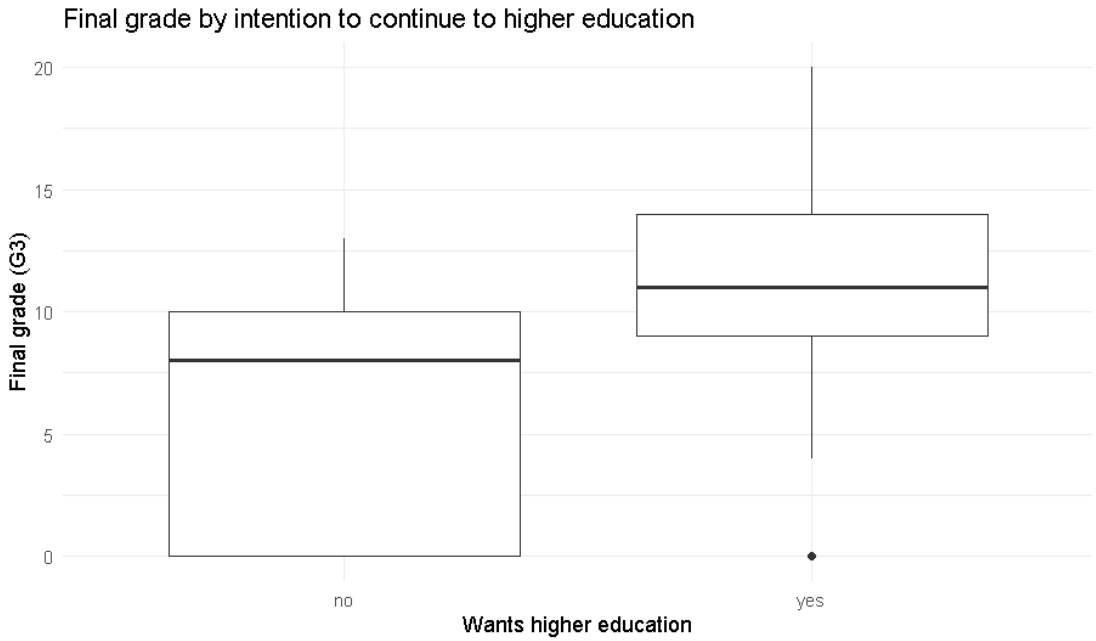
Note. The histogram displays the distribution of final Mathematics grade among 395 students.

Figure 2.



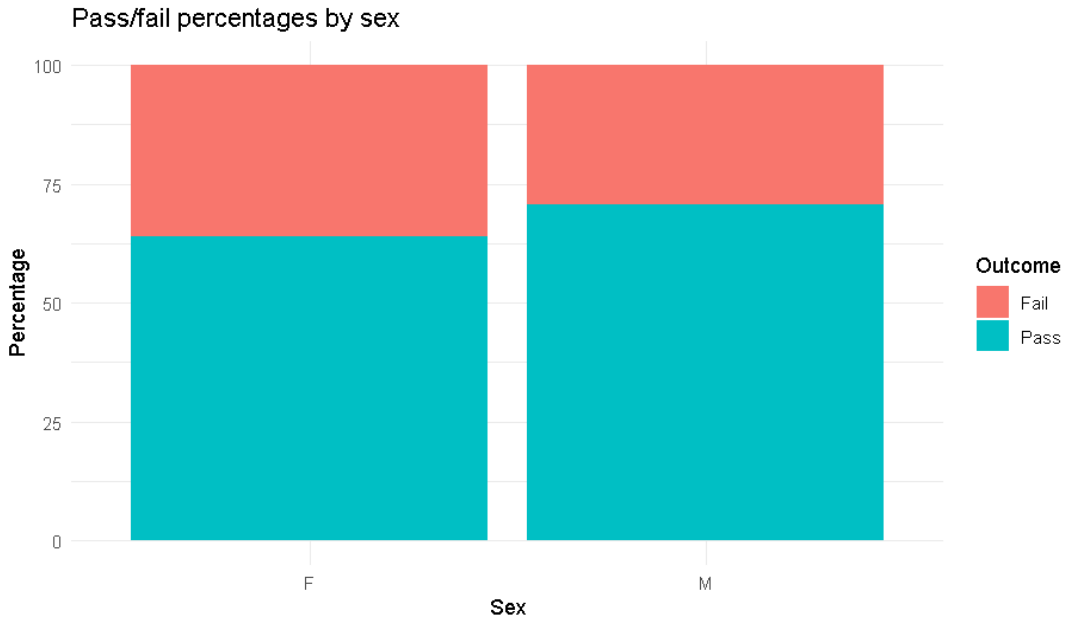
Note. The boxplot shows the distribution of final grade across four study-time categories: 1 (under 2 hours), 2 (2–5 hours), 3 (5–10 hours), 4 (over 10 hours)

Figure 3.



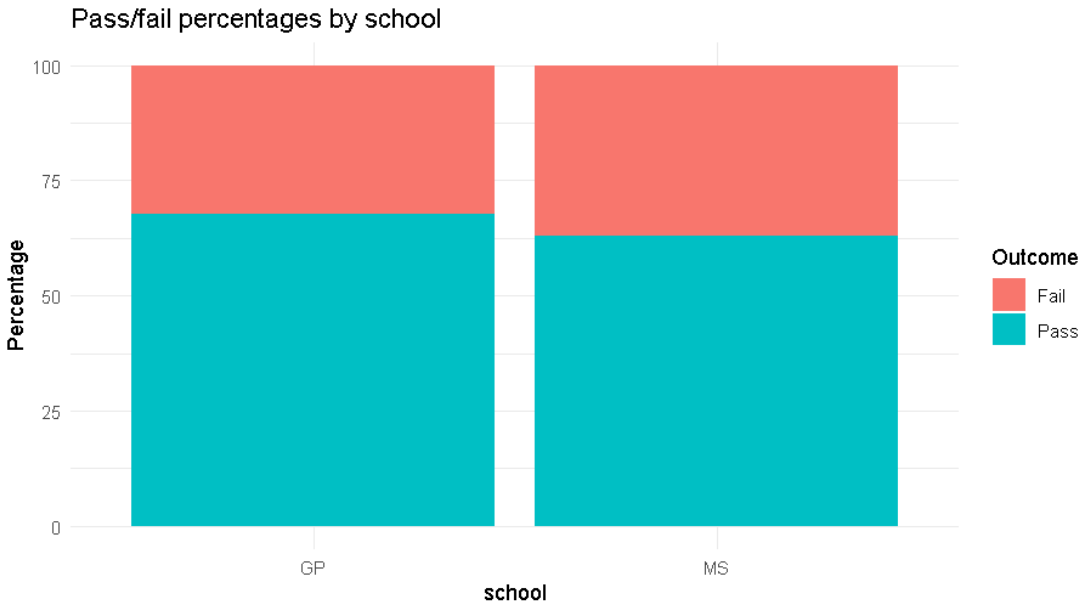
Note. This boxplot displays the distribution of final grade between students with a desire to pursue higher education and students without.

Figure 4.



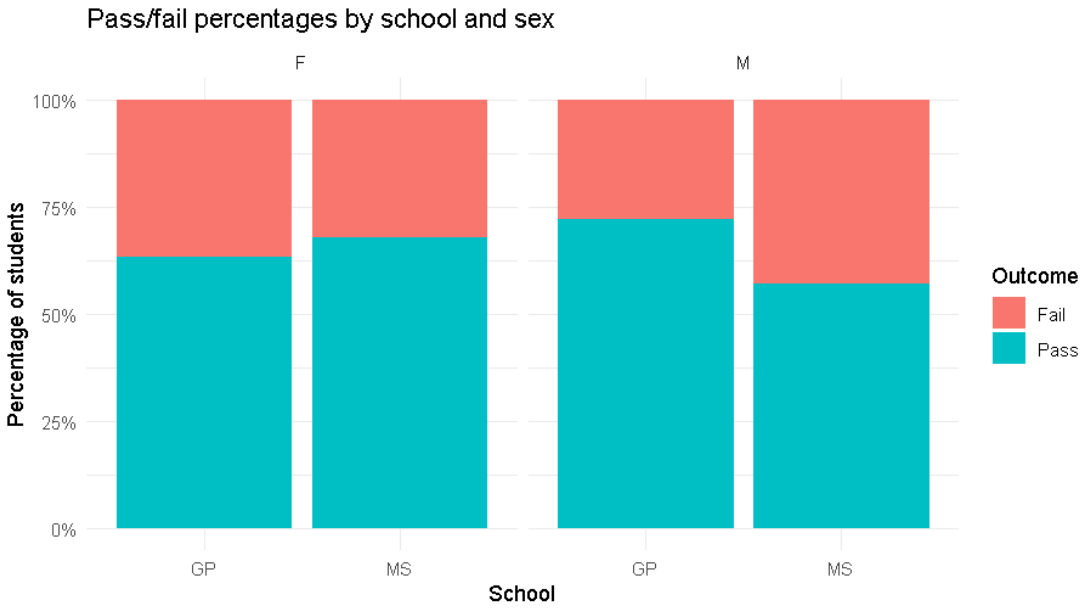
Note. This stacked bar chart compares percentages of students who passed and failed their Mathematics course by sex (F = Female, M = Male).

Figure 5.



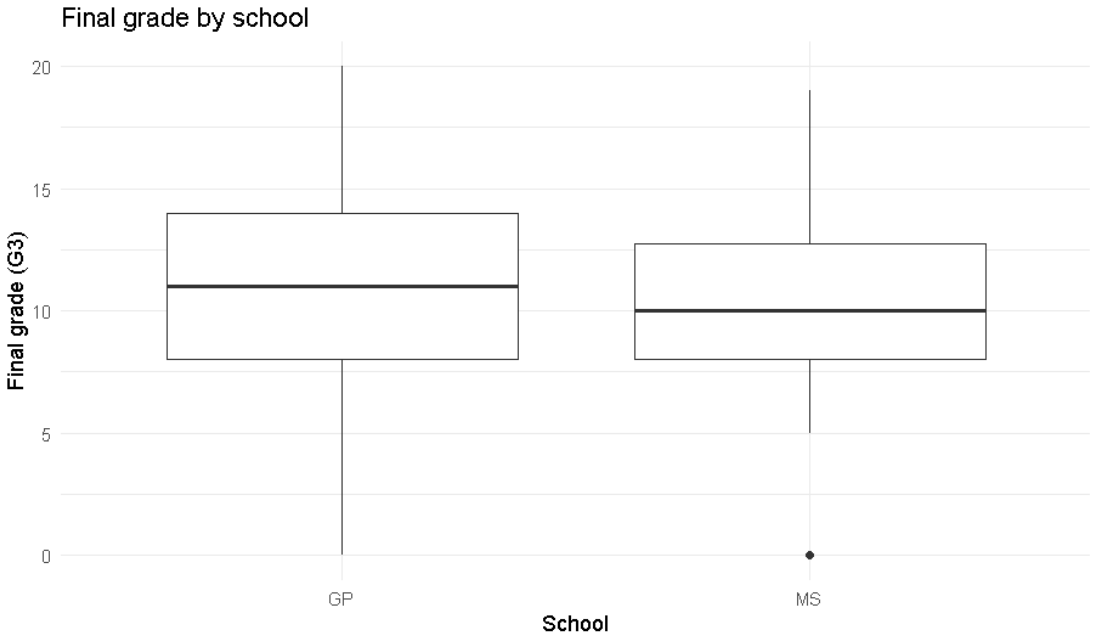
Note. This stacked bar chart compares percentages of students who passed and failed their Mathematics course by the school they attended (GP = Gabriel Pereira, MS = Mousinho da Silveira).

Figure 6.



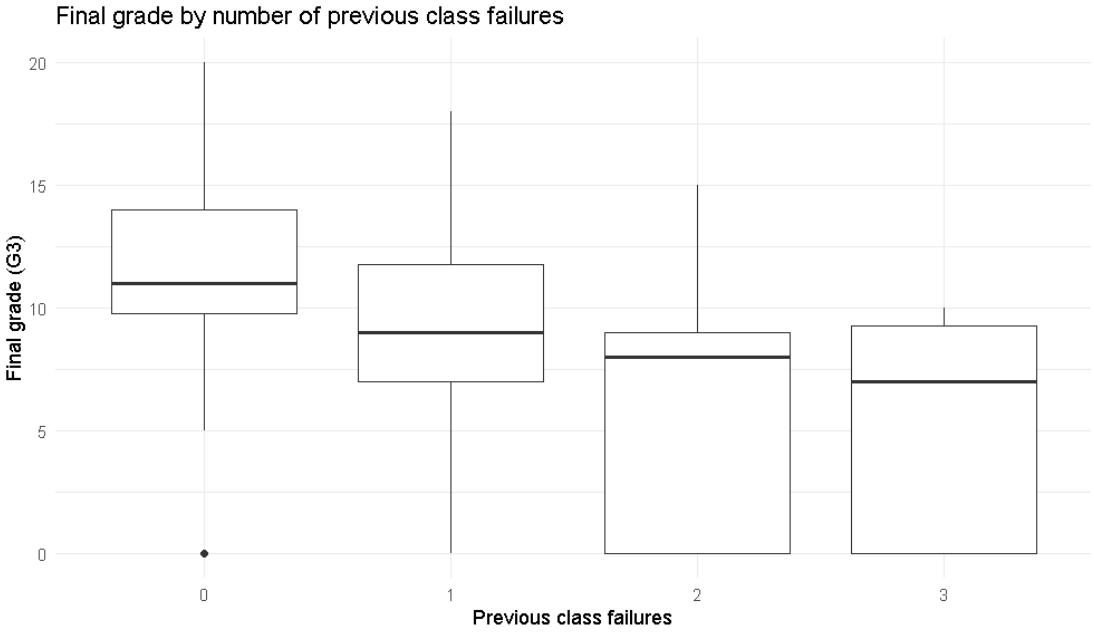
Note. This stacked bar chart compares percentages of female and male students who passed and failed their mathematics course at two school.

Figure 7.



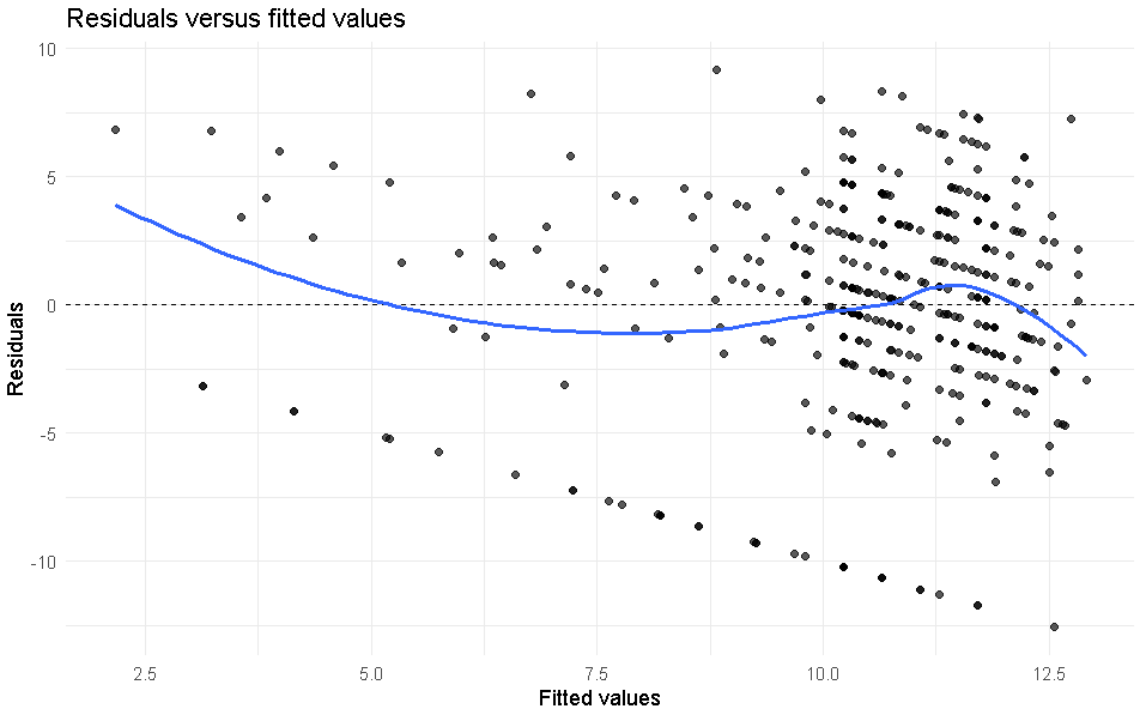
Note. This boxplot illustrates the distribution of final grades among students of two schools.

Figure 8.



Note. This boxplot displays the distribution of final grades and the number of past class failures.

Figure 9.



Note. The plot displays residuals against fitted values for the linear regression model, with a horizontal dashed line at zero and a blue smoothed line. The fitted values range from approximately 2 to 13, while the residuals range from approximately −13 to 9.